Bill of Materials

Automatic Chain-Stitch Crochet Machine (ESP32, Wi-Fi controlled)

All prices are approximate estimates in Indian Rupees including GST, intended for budgeting. They were not checked against live listings, so confirm price and stock before ordering. The purchase links open a search on the supplier site rather than a single product page, so they keep working if a listing changes.

No.	Item	Specification	Qty	Unit Price	Total	Purpose	Where to buy
1	ESP32 DevKit V1 (30-pin)	ESP32-WROOM-32, USB micro or Type-C, CP2102/CH340	1	Rs 450	Rs 450	Main controller, Wi-Fi web interface	Robu.in Amazon.in
2	NEMA 17 stepper motor	17HS4401, 1.8 deg, 1.5 A, 40 mm body	1	Rs 900	Rs 900	Hook vertical (Z) drive	Robu.in Amazon.in
3	A4988 stepper driver	With heatsink (TMC2209 is an optional quieter upgrade)	1	Rs 150	Rs 150	Drives the stepper motor	Robu.in Amazon.in
4	T8 lead screw with brass nut	8 mm diameter, 8 mm lead, 200 mm length	1	Rs 450	Rs 450	Converts rotation to hook up/down travel	Robu.in Amazon.in
5	Linear rod and bearing kit	8 mm smooth rod x2 (200 mm) with LM8UU bearings x2	1	Rs 450	Rs 450	Guides and stabilises the hook carriage	Robu.in Amazon.in
6	Flexible shaft coupler	5 mm to 8 mm aluminium	1	Rs 100	Rs 100	Joins motor shaft to lead screw	Robu.in Amazon.in
7	MG90S micro servo	Metal gear, 180 degree	2	Rs 200	Rs 400	Hook rotation and yarn arm	Robu.in Amazon.in
8	12 V DC power adapter	12 V, 3 A, 5.5 x 2.1 mm barrel plug	1	Rs 450	Rs 450	Main power supply	Robu.in Amazon.in
9	LM2596 buck converter module	Adjustable, set output to 5.0 V	1	Rs 150	Rs 150	Separate 5 V rail for servos	Robu.in Amazon.in
10	Electrolytic capacitor	470 uF, 25 V	2	Rs 20	Rs 40	Motor supply and servo rail smoothing	Robu.in Amazon.in
11	Micro limit switch	Lever type, NO/NC/COM	2	Rs 60	Rs 120	Z-axis home position (one spare)	Robu.in Amazon.in
12	Steel crochet hook	2.5 mm, metal	2	Rs 80	Rs 160	Working hook (one spare)	Robu.in Amazon.in
13	Perfboard or breadboard	830-point breadboard or 7x9 cm perfboard	1	Rs 150	Rs 150	Circuit assembly	Robu.in Amazon.in

No.	Item	Specification	Qty	Unit Price	Total	Purpose	Where to buy
14	Jumper wire set	Male-male and male-female, 40 pcs each	1	Rs 150	Rs 150	Connections	Robu.in Amazon.in
15	DC barrel jack terminal	5.5 x 2.1 mm screw-terminal adapter	1	Rs 50	Rs 50	Safe 12 V input connection	Robu.in Amazon.in
16	Fastener kit	Assorted M3, M4, M5 screws, nuts and washers	1	Rs 300	Rs 300	Frame and bracket assembly	Robu.in Amazon.in
17	Base and frame material	6 mm acrylic or 12 mm MDF base, about 300 x 200 mm	1	Rs 300	Rs 300	Mounting base	Robu.in Amazon.in
18	3D-printed brackets (print service)	PLA or PETG, hook holder, arm mount, motor mount	1	Rs 600	Rs 600	Custom mechanical parts	Robu.in Amazon.in
19	Cotton yarn	4-ply cotton, light colour for easier visibility	1	Rs 100	Rs 100	Test material	Robu.in Amazon.in
				Subtotal	**Rs 5,470**		
				Contingency (10%)	Rs 547		
				Estimated Grand Total	**Rs 6,017**		

Notes

- Tools not included: soldering iron, multimeter, screwdriver and hex key set, and access to a 3D printer or a print service.
- Servos must run from the separate 5 V buck converter, not from the ESP32 board.
- The A4988 can be replaced with a TMC2209 for quieter operation; wiring for STEP, DIR and EN stays the same.
- An editable version of this list with live formulas is provided as Crochet_Machine_BOM.xlsx.